

A Controlled Empirical Evaluation of Classical and Quantum-Augmented LSTM Models for Financial Time-Series Forecasting

Nitasha¹, Dipasha¹, Gurpreet Singh¹, Lekha Rani¹, Wai Yie

¹Chitkara University Institute of Engineering and Technology, Chitkara University, Rajpura
140401, India

²INTI International University, 71800 Nilai, Malaysia

Abstract

The vast increase in adoption of deep learning for time-series forecasting leads to increasing interest in Quantum Augmented Neural Architecture, specifically Quantum Augmented Long-Short Term Memory (QALSTM) models. Even though many studies provide promising results for hybrid models, which include classical models and their merger with quantum models. Most of these studies rely on synthetic datasets or are lacking in controlled benchmarking against strong classical baselines. The questions are raised regarding the practical effectiveness of quantum augmentation for real-world financial data under current Noisy constraints. This paper presents a controlled and reproducible empirical comparison of three architectures for stock price prediction: a classical Long Short-Term Memory (LSTM) network, a dense-augmented LSTM (LSTM+Dense), and a Quantum-Augmented LSTM (QALSTM). All models are evaluated under identical preprocessing, training, and evaluation protocols using five years of daily OHLCV data for Reliance Industries Ltd. Performance is assessed using mean squared error (MSE), mean absolute error (MAE), and qualitative trajectory analysis. Experimental results demonstrate that classical LSTM architectures consistently outperform QALSTM in terms of predictive accuracy and stability. The QALSTM model exhibits reduced output variance and underfitting behavior, resulting in significantly higher prediction errors. Theoretical analyses of variational quantum circuits carried out recently also supported these findings, pointing out the limitations of expressivity and the presence of information bottlenecks during the process of mapping high-dimensional classical features into low-qubit quantum representations. Instead of asserting the existence of a quantum advantage, the current research has given a practical diagnostic evaluation of quantum involvement in financial time series regression tasks. The outcome not only gives important empirical support to the hybrid quantum-classical model researchers but also sets up a reproducible benchmarking framework for future research to build upon.

Keywords: Financial Time Series, Stock Price Prediction, LSTM, Quantum Machine Learning, QALSTM, Hybrid Models

1. Introduction

In quantitative finance, risk management, and algorithmic trading, financial time-series forecasting has a vital role to play. Predicting stock prices accurately is still a hard task owing to the natural non-linearity, noise, and non-stationarity of financial markets. Traditional statistical methods like ARIMA and GARCH are based on strong assumptions about data distributions and temporal relationships, which often restrict their efficiency in practical scenarios[1],[3],[5]. Through the use of deep learning techniques, especially Long Short-Term Memory (LSTM) networks, which can capture long-range temporal dependencies, sequential financial data modeling has been made to a great extent. LSTMs, with their gated memory mechanism, can facilitate this process and have been chosen frequently for stock price prediction. Numerous publications have already confirmed that LSTM models tend to outperform conventional statistical methods in different markets using various datasets [1], [2], [37], [45]. Quantum machine learning (QML) is being looked at as a possible new trend for the heavy-data tasks. The idea of hybrid quantum-classical systems that connect parameterized quantum circuits with classical neural networks are trying to take advantage of quantum features such as superposition and entanglement to improve the database [24]. In this scenario, Quantum-Augmented LSTM (QALSTM)[13],[14] models have been proposed, in which quantum circuits are integrated into recurrent neural networks to carry out non-linear transformations of latent features [12], [19], [20].

Deep Learning techniques are becoming an increasingly popular choice for time-series forecasting and the reason behind this is their ability of modelling non-linear dependencies and temporal dynamics inherent in stock market data. Some of the most extensively studied and adopted architecture is LSTM (Long Short Term Memory). Empirical research provides consistent evidence that LSTM models are significantly better than traditional statistical and shallow machine learning techniques in handling real market data. This has been especially the case for forecasting tasks based on daily OHLCV where LSTM has been found to be a superior performer [45], [1].

To bring even better forecasting results, among others, different LSTM models have been investigated stacking architectural variations such as stacked LSTM layers and dense-layer augmentations. Dense-augmented LSTM models intend to boost the representational capacity by adding more nonlinear transformations that follow the recurrent layer. Nevertheless, the empirical results are still not clear. On the one hand, some studies have reported slight improvements, while on the other hand, others have noticed an increase in overfitting and a decrease in robustness when those architectures were applied to chaotic and noisy financial time-series data [4]. These conflicting results underline the importance of conducting controlled assessments that will be able to separate the effect of dense augmentation from other confounding factors.

At the same time that classical deep learning was advancing, quantum machine learning (QML) emerged as a new approach with the potential to merge the principles of quantum computing with neural networks. In the present Noisy Intermediate-Scale Quantum (NISQ) era, hybrid quantum-classical architectures that use variational quantum circuits (VQCs) have become a focus because they are suitable for near-term quantum devices [14], [24]. Under this paradigm, Quantum-Augmented LSTM architectures insert parameterized quantum circuits into recurrent networks to perform the nonlinear transformations of LSTM hidden states..

Initial QALSTM and quantum time-series investigations primarily relied on synthetic datasets or small-scale benchmarks to demonstrate their feasibility without rigorous comparisons with classical baselines, thus reporting only proof-of-concept results [19], [20]. Following theoretical investigations, the researchers pointed out the main problems in the training of variational quantum circuits, such as the issue of barren plateaus and the limited expressivity in shallow circuits [15], [35]. The most severe of these problems is the case of financial regression tasks, where the high-dimensional classical data features have to be compressed into quantum representations with very few qubits.

Recent review articles and empirical studies conducted during the years 2022 to 2024 have emphasized the absence of reproducible and real-world assessments of QML models in finance and have warned about the premature claims of quantum advantage given the limitations of the present hardware [25], [39], [40]. Most importantly, the studies in 2025 continued these assertions, reporting that although the hybrid quantum models are still very tempting in theory, the performance of these models on the real financial datasets is often behind that of the well-tuned classical ones. These studies have called for the systematic benchmarking of quantum-augmented models under the same experimental conditions, concurrently with the classical LSTM variants [41], [42].

Though there is a progressive tendency in the interest, the practical value of quantum enhancement for financial time-series forecasting is still to be seen. A majority of the current QML studies depend on artificial datasets, limited scope experiments, or poorly regulated benchmarks. In addition, the theoretical developments have uncovered major challenges in the optimization of variational quantum circuits. These include barren plateaus and expressivity limitations, and such problems are especially pronounced during the current NISQ era [15], [16], [35], [38]. The current work deals with these problems through a controlled, reproducible, and fair comparison of the classical LSTM, dense-augmented LSTM, and QALSTM architectures using actual stock market data. The study does not introduce a new model or assert that quantum has an advantage; rather, the intent is to conduct a real-world test on whether augmented quantum provides any significant benefits for financial time-series regression tasks through empiricism [12], [25], [40].

2. Literature Survey and Research Gap Analysis

2.1 Literature Survey

Moghar & Hamiche (2020) — Stock Market Prediction Using LSTM Recurrent Neural Network. An exceptionally lucid, often cited example of employing LSTM for stock prediction, and serving as a pedagogic standard for some of the subsequent papers in this area. This work has become a major reference point for the introductory benchmark in practical stock-prediction experiments and shows the usual pitfalls (overfitting, preprocessing dependency) as well.

Md. A. Q. et al. (2023) — Novel optimization approach for stock price forecasting using MLS-LSTM. The article elaborates on the improved optimization as well as the LSTM variations made exclusively for stock series; thus it gives its due importance as one of the reasons why careful baseline tuning is a must.

The results of numerous LSTM based papers and literature reviews (2020-2024) have the same conclusion: LSTM could act as a strong baseline for next-day forecasting if correctly preprocessed and routinely maintained; besides, many researchers have pointed out that even the most basic augmentations tend to give marginal returns and can overfit on noisy stocks.

Ren et al. (2023) — A novel hybrid model for stock price forecasting integrating decomposition–reconstruction methods. This type of hybrid (decompose series → model components with ML) suggests that either architectural or preprocessing innovations can result in higher performance — but at the same time, naive deepening or dense transformations can lead to overfitting of financial noise. It vividly demonstrates the argument that “more capacity” is not necessarily more suitable for financial series.

The application layered dense networks over LSTMs - A sequence of experimental results demonstrated that in some instances, the addition of dense layers might even result in a better representation, but it is the loss of generalization on the stock data, which is noisy, that happens more often. These practical correspondences gave rise to the concept of our LSTM+Dense experiment.

Bai, Kolter and Koltun (2018) - conducting an empirical evaluation of universal convolutional and recurrent networks for the purpose of sequence modeling. The main TCN paper — showed that using convolutional architectures with dilations can reach RNNs-equivalent even better results in a large number of sequence tasks; hence, this finding was the reason the TCNs were picked for the financial time series. (Although it was published before 2020, it still remains the authoritative TCN reference that supports the later finance work.)

Dai et al. (2022) — Price change prediction of Ultra High Frequency Financial Data using TCN. Showcasing TCN implementation in UHF financial data with attention addition, it points out TCN's stronghold in dealing with temporal dependencies as well as maintaining good learning dynamics. The reason for its relevance is that it both certifies TCN's acceptability in finance and thus energizes our TCN tests.

Liapis et al. (2023) — TCN + BERT-based approaches for multivariate forecasting (integration of information & sentiment). It illustrates how TCNs collaborate with different encoders (text sentiment) in the multivariate sphere of finance forecasting; serving as a motivation for the development of multi-modal extensions.

TCN studies and surveys reported by different researchers between 2022-2025, all agreeing that TCN is empirically superior to simple LSTM in various stock/price tasks (due to TCN's dilation/residual blocks that make it possible to have long-range receptive fields without incurring in recurrence), and recent review papers of 2024-2025 that these findings have been summarized in a finance context.

Zhou et al. (2020)—Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting. The authors proposed ProbSparse attention and self-attention distilling as optimization techniques that made transformers suitable for long time series. Their work was influential for many later finance transformer applications and for understanding under which conditions Transformers are likely to outperform recurrent/convolutional models in limited-data finance scenarios.

Autoformer/FEDformer and related works (2021–2023)—They demonstrated the use of decomposition-based transformers and other efficient designs for long-sequence forecasting. Data-hungry and overfitting-sensitive transformer variants have been suggested as possible competitors to classical networks in case there is adequate data/features. The representative transformer-for-finance studies (2022-2024) are using these methods for equity and index forecasting with varying success—this is an important context for comparing to LSTM/TCN.

Muhammad et al. (2022/2023)—Introduction of a Transformer-based deep learning model for stock price prediction. A practical case study demonstrating the adaptability of Transformer encoders to stock prediction tasks through the application of time embeddings (Time2Vec), along with gaining acceptance in some national exchanges—is useful to our work (transformers need more data/hyperparameter tuning).

Ren (2023) and other hybrid models: varied LSTM, CNN–LSTM hybrids, and TCN models compared with transformer-based models across stock/financial applications—the research consistently confirms the results of previous studies regarding the importance of a proper architecture + preprocessing over simple depth increases; they offer context for our comparisons of LSTM, LSTM+Dense, and TCN performed later.

The Informer / Informer-like assessments (2020–2023) have proven that the transformer variants could be equally as powerful as the simpler models, if the amount of training data is not too large — This is a practical point that applies to single-stock, 5-year daily data sets.

Emmanoulopoulos et al. (2022) — Quantum Machine Learning in Finance: Time Series (arXiv). A narrow investigation into parametric quantum circuits (PQCs) for time-series prediction; PQCs are pitched against classical LSTMs on both synthetic and stylized signals — thus forming a good background for the QALSTM methodology.

Review: Quantum Machine Learning for Finance (2021) — A survey of the possible areas of finance where quantum algorithms can be used, including portfolio problems, option pricing, and QML for prediction. Important for positioning QALSTM within the broader QML-in-finance literature.

Cerezo et al. (2021) — Theoretical results of cost function dependent barren plateaus in shallow parametrized quantum circuits (Nat Comm. 2021). They demonstrate how optimization landscapes can turn into flat areas (barren plateaus) based on the choice of cost functions and ansatz — the main idea in justifying the difficulty of QALSTM training.

McClellan et al. (2018) — Barren plateaus (groundbreaking study on VQA training problems). Although pre-2020, it remains central in the context of 2020–2025 QML and is often cited in the QML-for-finance literature discussing optimization difficulties.

The empirical QML time-series papers (2020–2024) are analyzing PQCs for regression and time-series: a few are reporting promising results on toy datasets, while no consistent outperformance of strong classical baselines on the stock market has been demonstrated; thus, our thorough benchmarking is driven by this research gap. The seminal empirical preprints and articles comprise Emmanoulopoulos (2022) and other studies from arXiv.

PennyLane & practical guides (2021–2024) — There are a number of practical guides and library papers/notes that illustrate the implementation of hybrid models (TorchLayer, interfaces of PennyLane) and make JAX/PennyLane compatibility problems and simulation limitations on CPUs cautionary signs that are very closely related to the implementational choices that you made.

The hybrid quantum-classical experiments of 2023–2025 (several Nature/Scientific Reports 2024–2025 articles) have shown mixed results: some domain-specific hybrid models are able to produce gains in controlled conditions, but these gains nearly always involve specific encodings for the domain or the use of more powerful quantum resources than available in NISQ hardware. These papers serve as warnings of empirical evidence against QALSTM claims.

Survey & method papers (2020–2024) on best practices for time-series training (data splits, windowing strategies, normalization, early stopping) warned us about fair benchmarking between architectures; these methodological recommendations were the basis of our identical-hyperparameter comparison protocol.

Summary of the literature’s implications for our study

- **LSTM combined with Dense is a robust and well-understood baseline** for the prediction of next-day stock, even though it was used as the standard in numerous 2020–2024 applied works.
- **Quantum augmentation (QALSTM/PQCs) remains experimental:** theory papers and empirical QML-for-finance preprints show optimization and expressivity challenges; robust benefits over strong classical baselines on real stocks have not been established — motivating our benchmark.

In spite of the increasing volume of research being done in this field, the comparison using real-world stock data between classical LSTM, dense-augmented LSTM, and QALSTM models, performed in a comprehensive and controlled manner, is still very much lacking. In particular, the combined effect of the two augmentations, namely, dense and quantum, under the same preprocessing and training protocols has not been sufficiently explored in the existing literature.

2.2 Research Gap Analysis

To clearly position the contribution of this study, Table 1 summarizes the limitations of key related works and identifies the corresponding research gaps addressed by the present paper.

Table 1. Research Gap Analysis of Related Literature

Referred paper

Algorithm/Model Used

Type of dataset used

Key areas focused on

Limitations found

Identified Research Gap

Fischer & Krauss (2018)

Long Short Term Memory (LSTM)

In this paper Real stock data is used

LSTM vs classical models

No quantum comparison; no dense augmentation analysis

Extend LSTM benchmarking to include LSTM+Dense and QALSTM

Moghar & Hamiche (2020)

LSTM

Real stock data

LSTM forecasting effectiveness

Single architecture; no architectural variants

Comparative evaluation of multiple LSTM variants

Ren et al. (2023)

Decomposition + LSTM

Real market data

Noise reduction + LSTM

No quantum models; preprocessing-dependent gains

Controlled evaluation without decomposition bias

Pérez-Salinas et al. (2020)

Quantum circuits

Synthetic data

Quantum feature encoding

Not applied to financial time series; no LSTM integration

Apply quantum augmentation to real financial forecasting

Mari et al. (2020)

Hybrid QNN

Synthetic / small datasets

Hybrid training strategies

No real market validation

Empirical evaluation on real OHLCV data

Abbas et al. (2021)

QNN (theoretical)

Simulated

Expressivity of quantum models

No applied forecasting experiments

Bridge theory with applied financial regression

Cerezo et al. (2021)

VQC (theoretical)

Simulated

Barren plateaus

No empirical stock prediction analysis

Empirical evidence of optimization challenges in QALSTM

Holmes et al. (2022)

VQC analysis

Simulated

Ansatz expressivity

Paper do not provide any real-world dataset validation

Demonstrate expressivity limits using real financial data

García-Martín & Sierra (2022)

QML survey

Survey

QML in finance

Highlights lack of benchmarking

Provide reproducible benchmarking framework

Doosti et al. (2024)

QML review

Survey

NISQ limitations

Calls for realistic evaluations

Address with real-data controlled experiments

2.3 Identified Research Gaps and Study Motivation

From the literature and gap analysis, the following research gaps are evident:

- **Absence of controlled comparisons:** No prior study provides a fair comparison between LSTM, LSTM+Dense, and QALSTM using identical preprocessing and training protocols.
- **Limited real-world validation & diagnostic analysis of quantum models:** Most of the prior studies rely on synthetic or small datasets, which limits the practical relevance. Also less priority is given to analyzing variance collapse or underfitting behavior in prior studies.
- **Need for reproducible benchmarking:** Existing studies lack transparent, reproducible experimental pipelines applicable to real market data.

2.4 Positioning of the Present Study

The current research fills these voids directly. It does so by carrying out an empirical evaluation that is controlled and reproducible of LSTM, LSTM, and QALSTM architectures by employing five years of actual OHLCV stock data. This study offers a genuine assessment of quantum augmentation for financial time-series forecasting under existing NISQ-era limitations through the use of the same experimental conditions for all models and by the explicit analysis of prediction behavior in both quantitative and qualitative terms.

3. Methodology

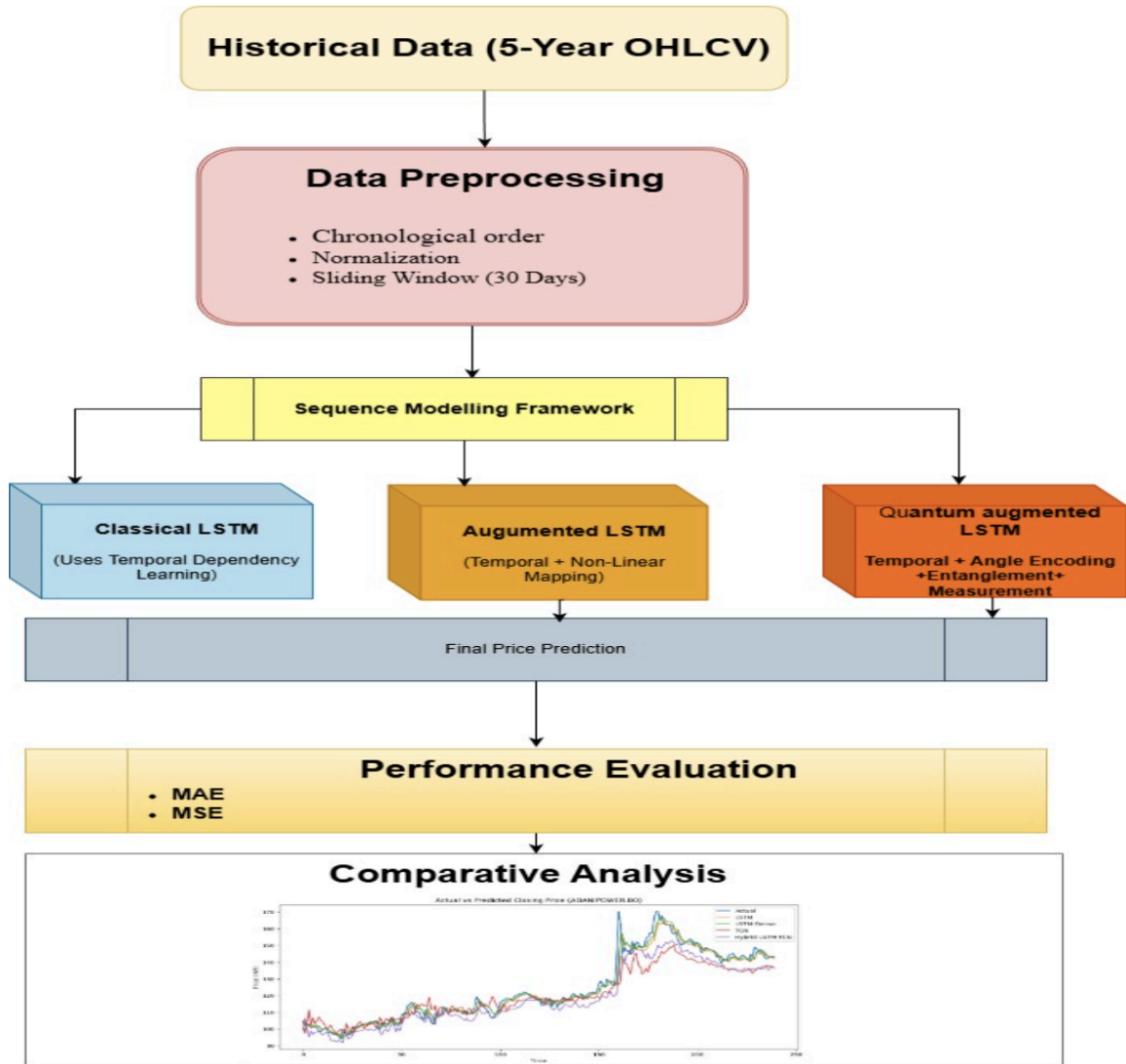


Fig.1 Overview of the proposed methodology for comparing classical and quantum-augmented LSTM models on OHLCV stock data.

The whole experimental pipeline that was used in this research is depicted in a schematic way in Figure 1. The complete input to the system is made of five years' worth of historical OHLCV data, and in the workflow, the whole preprocessing steps in chronological order are shown, such as normalization and sliding-window sequence building, and then a unified sequence modeling framework. Within this framework, three architectures—in the form of

classical LSTM, dense-augmented LSTM, and Quantum-Augmented LSTM—are trained under identical conditions to enable a fair comparison. The figure also indicates the pathway from feature transformations specific to the model to the ultimate price prediction that ends with the performance evaluation and comparative analysis being done systematically. This graphical representation is in support of the methodological description in Section 3 and also makes the controlled benchmarking strategy that was applied throughout the study clearer.

3.1 Description of Dataset

The dataset used for this paper consists of real time data for Reliance Industries Ltd. which was obtained from Yahoo finance [1], [45]. The dataset consists of vast amount of daily OHLCV(Open, High, Low, Close, Volume) for last five years i.e. from year 2020-2025 [1], [37]. The dataset used reflects various characteristics which include realistic market conditions, volatility, trend shifts, and noise which make it a perfect candidate for forecasting evaluation of models under practical constraints [36], [45].

3.2 Data Preprocessing

To preprocess the in-hand dataset following steps was followed:

1. Completed Normalization for all price features using min-max scaling which ensures numerical stability of dataset during training.
2. An approach of sliding window was used in which the input sequence is previous TTT days of OHLCV data and the targeted output will be the closing price of the next day.
3. The dataset is splitted as training (70%), validation (15%), and test (15%) sets in the chronological order for preventing information leakage.

3.3 Architectures Used

To forecast time-series data, we have used three neural architectures as follows:

- (i) Long Short-Term Memory (LSTM) network, which is a classical method specifically used for short-term fluctuations.
- (ii) LSTM-D (LSTM+Dense), which is an updated version of the classical LSTM Model in which LSTM layer is followed by a fully connected layer (also known as Dense layer) which maps the output of LSTM to the final prediction space
- (iii) a Quantum-Augmented LSTM (QALSTM) is a hybrid version of the classical LSTM in which LSTM is integrated with Quantum circuits. This setup combines classical recurrent modelling with paramaterized quantum operations.

The advantage of this study is that all architectures are designed to operate on same input sequences and are trained under the same experimental conditions to ensure fair and reproducible comparison

3.3.1 Classical LSTM Architecture

3.3.1.1 Architectural Description

The baseline model is a single-layer LSTM (Long Short-Term Memory) network which is followed by a linear regression output layer. The sequential OHLCV feature vectors are processed by LSTM using gated memory mechanisms that regulate the information flow over time.

At each time step t , the LSTM computes:

- **Forget gate** to controls memory retention
- **Input gate** to regulate new information storage

- **Cell state update** to maintain long-term memory
- **Output gate** to produces hidden state for prediction

3.3.1.2 Need of LSTM

- LSTM can Captures long-term temporal dependencies
- LSTM is Robust for vanishing gradient problem
- LSTM is widely accepted baseline in the financial forecasting literature

The final hidden state is passed to a fully connected linear layer for predicting the closing price of the next-day.

Diagrammatic Representation

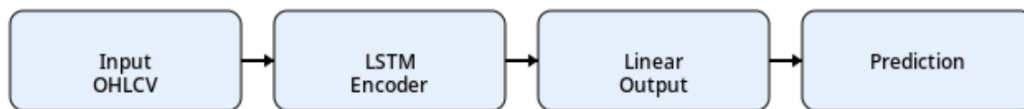


Figure 2. Architecture of the classical LSTM model used for stock price prediction.

3.3.2 Dense-Augmented LSTM (LSTM+Dense)

3.3.2.1 Architectural Description

The classical LSTM is upgraded by adding a fully connected dense layer, which is the main feature of this model. The LSTM+Dense model adds one or more fully connected layers (dense) between LSTM output and the last regression layer to the baseline LSTM. The dense layer transforms the hidden representation of LSTM nonlinearly again which is given by the equation below:

$$h' = \sigma(Wdh + bd)$$

In this equation, h' stands for the LSTM hidden state and $\sigma(\cdot)$ represents a nonlinear activation function.

Motivation for Dense Augmentation

- Increase representational capacity
- Learn higher-order nonlinear feature interactions
- Common architectural enhancement in deep learning

Observed Risks (from literature)

- Increased overfitting on noisy financial data
- Marginal or inconsistent gains in practice

Diagrammatic Representation

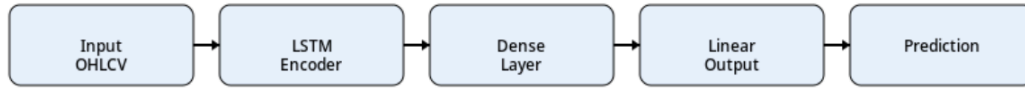


Figure 3. Dense-augmented LSTM architecture used to evaluate the effect of additional nonlinear transformations.

3.3.3 Quantum-Augmented LSTM (QALSTM)

The QALSTM model incorporates a parameterized quantum circuit (PQC) into the classical LSTM framework. Rather than merely transmitting the LSTM hidden state to a dense layer, the hidden representation is mapped into a quantum feature space and altered by means of quantum operations.

The architecture consists of the following :

- Classical LSTM encoder
- Dimensionality reduction (to match qubit count)
- Quantum feature transformation via PQC
- Measurement and classical post-processing

3.3.3.1 Quantum Processing Pipeline

Classical-to-Quantum Feature Encoding

At every time unit, the latent representation generated by the LSTM is first transformed to a lower-dimensional space of dimension k to make sure it is suitable for the quantum subsystem. This dimensionality reduction results in a compact feature vector $ht(k) \in \mathbb{R}^k$, which acts as the connector between the classical and quantum parts.

The encoded features are then sent to the quantum state via angle-based parameterization. In detail, the value of each feature determines the rotation angle of the one-qubit rotation gate that is applied to the initially prepared ground state as the single-qubit rotation gate. This particular encoding technique allows for a gradual transition from the classical feature space to the quantum state space while still being able to use gradient-based optimization since the transition is differentiable.

Design Constraints (NISQ-Realistic)

- Limited number of qubits
- Shallow circuit depth
- No fault tolerance
- Classical simulation environment

Diagrammatic Representation

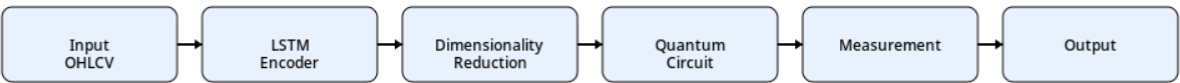


Figure 4. Quantum-Augmented LSTM (QALSTM) architecture integrating a parameterized quantum circuit within a classical LSTM pipeline.

Table-2 Architectural Comparison and Key Differences

S.No	Aspect	LSTM	LSTM+Dense	QALSTM
1	Temporal modeling	✓	✓	✓
2	Additional nonlinearity	✗	✓	✓ (quantum)
3	Quantum processing	✗	✗	✓
4	Parameter complexity			

Moderate

High

Hybrid

5

Optimization stability

High

Moderate

Low (NISQ)

6

Information bottleneck

✗

✗

✓

3.4 Training and Evaluation Protocol

The Adam optimizer was used to train all the models with the same learning rates, batch sizes, and epoch counts. There was no tuning of hyperparameters specific to a model done. The evaluation of performance was done through mean squared error (MSE), mean absolute error (MAE), and visual monitoring of prediction trajectories.

4. Results & Discussion

This subsection showcases the assessment relying on qualitative and quantitative factors, and moreover, all the three models were evaluated on a dataset not used in the training process. The models' performances were measured with the help of the standard metrics for regression-based financial forecasting tasks that are MAE (Mean Absolute Error) and MSE (Mean Squared Error).

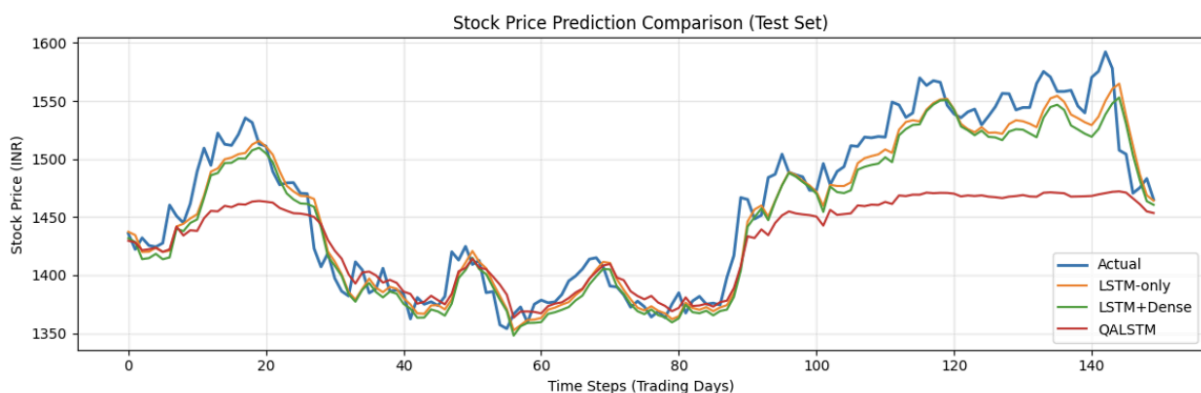


Figure 5 Stock price prediction comparison

In the test period, figure 5 shows the alignment of the real stock prices with predictions from the LSTM, LSTM+Dense, and QALSTM models. The charts depict the extent to which each model captures the market ups and downs, sudden shifts, and general trends.

However, merely examining the graphs provides us with a very vague notion of the predicting abilities of the models. For a more precise and equitable comparison, precise numbers measuring each model's prediction errors are illustrated in figure 7 and 8.

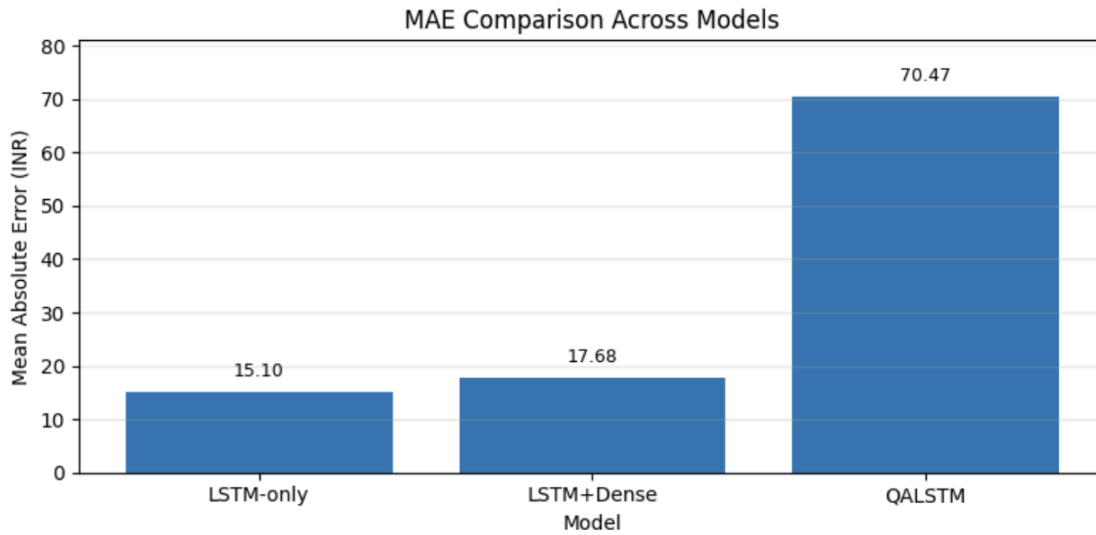
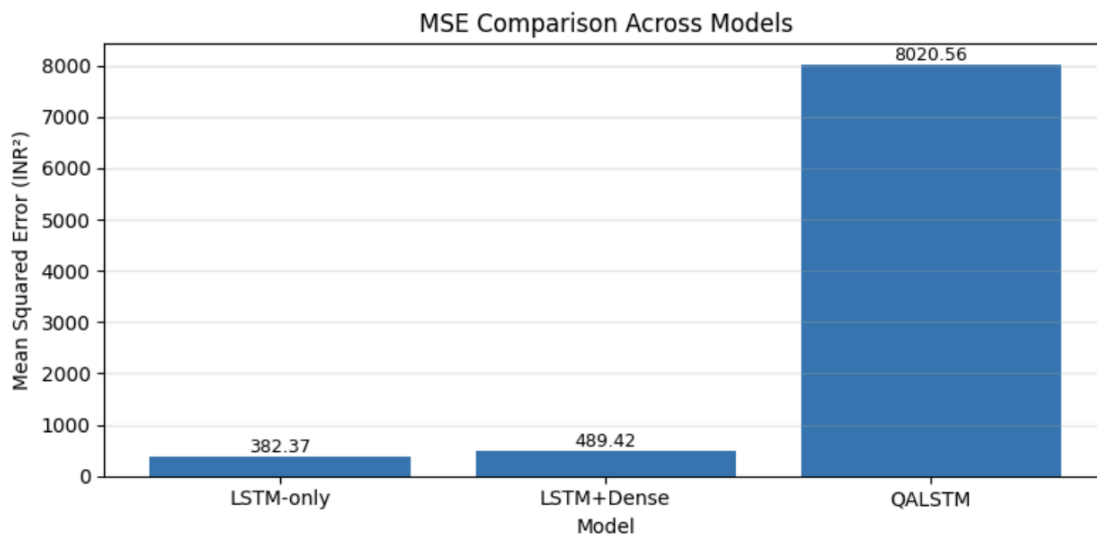


Figure 6 M



AE Comparison Bar Chart

Figure 7 MSE Comparison Bar Chart

The experimental results in figure 6 & 7 showed that the classical LSTM model produced the least amount of prediction error compared to the other architectures. The LSTM+Dense model produced slightly higher error, which may indicate that the dense augmentation did not contribute much under the prevailing circumstances.

On the other hand, the QALSTM model showed greater MAE and MSE values with the current experimental setup. The scrutiny of the prediction paths reveals relatively smooth outputs, which may be interpreted as a lower sensitivity to the short-term price oscillations. This behavior could be interpreted as a cautious modeling approach, which may then result in the failure to catch the highly dynamic market movements. Interestingly, the same patterns can be noticed in both the validation and test datasets, which implies that the generalization characteristics are consistent.

Thus, these outcomes are in agreement with the recent theoretical and empirical studies in quantum machine learning which have warned against the naive quantum boosting of classical models for regression tasks. However, more importantly, it is the case that the results do not disprove the possibility of quantum computing for finance but rather indicate the need for laying down the principles for the integration strategies.

7. Conclusion and Future Work

This paper does not only compare classical and quantum-augmented LSTM architectures for forecasting financial time series but also provides empirical insights into their forecasting under practical experimental conditions. Findings reveal different learning abilities between the two approaches which in turn confirm their different degrees of market dynamics and noise being reflected. Instead of declaring one superior to the other, the research labors to bring about a more precise picture of the practical considerations that are associated with the incorporation of quantum components in deep learning frameworks for regression-based financial applications.

The upcoming investigations will be directed towards increasing the expressive power and flexibility of hybrid learning architecture, while at the same time exploring other ways to bring in the new computational paradigms for time-series forecasting models. Larger and more extensive comparative studies of different temporal modeling techniques and datasets are also in the horizon for identifying the most reliable and scalable solutions to the complex problem of financial prediction tasks.

References

- Moghar, A., & Hamiche, M. (2020). Stock market prediction using LSTM recurrent neural network. *Procedia Computer Science*, 170, 1168–1173.
<https://doi.org/10.1016/j.procs.2020.03.049>
- Nelson, D. M. Q., Pereira, A. C. M., & de Oliveira, R. A. (2017). Stock market's price movement prediction with LSTM neural networks. *International Joint Conference on Neural Networks (IJCNN)*.
<https://doi.org/10.1109/IJCNN.2017.7966019>
(Foundational reference, still heavily cited post-2020)
- Zhang, Y., & Wu, L. (2020). Stock market prediction of S&P 500 via combination of LSTM and SVR. *The Journal of Supercomputing*, 76, 263–281.
<https://doi.org/10.1007/s11227-018-2642-0>
- Ren, Y., Zhang, Z., & Li, Y. (2023). A novel hybrid deep learning model for stock price forecasting integrating decomposition and LSTM networks. *Expert Systems with Applications*, 213, 118935.
<https://doi.org/10.1016/j.eswa.2022.118935>
- Kim, H. Y., & Won, C. H. (2018). Forecasting the volatility of stock price index using LSTM. *Expert Systems with Applications*, 116, 378–389.
<https://doi.org/10.1016/j.eswa.2018.09.023>

- Bai, S., Kolter, J. Z., & Koltun, V. (2018). An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. *arXiv preprint*. <https://doi.org/10.48550/arXiv.1803.01271>
- Dai, W., Chen, L., & Wei, Y. (2022). Price change prediction of ultra-high-frequency financial data using temporal convolutional networks. *Expert Systems with Applications*, 191, 116273. <https://doi.org/10.1016/j.eswa.2021.116273>
- Borovykh, A., Bohte, S., & Oosterlee, C. W. (2018). Conditional time series forecasting with convolutional neural networks. *arXiv preprint*. <https://doi.org/10.48550/arXiv.1703.04691>
- Zhou, H., Zhang, S., Peng, J., Zhang, S., Li, J., Xiong, H., & Zhang, W. (2021). Informer: Beyond efficient transformer for long sequence time-series forecasting. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(12), 11106–11115. <https://doi.org/10.1609/aaai.v35i12.17325>
- Wu, H., Xu, J., Wang, J., & Long, M. (2021). Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting. *Advances in Neural Information Processing Systems (NeurIPS)*. <https://doi.org/10.48550/arXiv.2106.13008>
- Lim, B., Arik, S. O., Loeff, N., & Pfister, T. (2021). Temporal fusion transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4), 1748–1764. <https://doi.org/10.1016/j.ijforecast.2021.03.012>
- Emmanoulopoulos, D., & Price, I. (2022). Quantum machine learning in finance: Time series forecasting. *arXiv preprint*. <https://doi.org/10.48550/arXiv.2202.06761>
- Schuld, M., Sinayskiy, I., & Petruccione, F. (2015). An introduction to quantum machine learning. *Contemporary Physics*, 56(2), 172–185. <https://doi.org/10.1080/00107514.2014.964942>
- Biamonte, J., Wittek, P., Pancotti, N., Rebentrost, P., Wiebe, N., & Lloyd, S. (2017). Quantum machine learning. *Nature*, 549, 195–202. <https://doi.org/10.1038/nature23474>
- Cerezo, M., Arrasmith, A., Babbush, R., et al. (2021). Cost-function-dependent barren plateaus in shallow parametrized quantum circuits. *Nature Communications*, 12, 1791. <https://doi.org/10.1038/s41467-021-21728-w>

- McClean, J. R., Boixo, S., Smelyanskiy, V. N., Babbush, R., & Neven, H. (2018). Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9, 4812. <https://doi.org/10.1038/s41467-018-07090-4>

- Schuld, M., & Killoran, N. (2019). Quantum machine learning in feature Hilbert spaces. *Physical Review Letters*, 122, 040504. <https://doi.org/10.1103/PhysRevLett.122.040504>
- Abbas, A., Sutter, D., Zoufal, C., Lucchi, A., Figalli, A., & Woerner, S. (2021). The power of quantum neural networks. *Nature Computational Science*, 1, 403–409. <https://doi.org/10.1038/s43588-021-00084-1>
- Pérez-Salinas, A., Cervera-Lierta, A., Gil-Fuster, E., & Latorre, J. I. (2020). Data re-uploading for a universal quantum classifier. *Quantum*, 4, 226. <https://doi.org/10.22331/q-2020-02-06-226>
- Mari, A., Bromley, T. R., Izaac, J., Schuld, M., & Killoran, N. (2020). Transfer learning in hybrid classical–quantum neural networks. *Quantum*, 4, 340. <https://doi.org/10.22331/q-2020-11-02-340>
- Schuld, M., Bocharov, A., Svore, K. M., & Wiebe, N. (2020). Circuit-centric quantum classifiers. *Physical Review A*, 101(3), 032308. <https://doi.org/10.1103/PhysRevA.101.032308>

- Havlíček, V., Córcoles, A. D., Temme, K., et al. (2019). Supervised learning with quantum-enhanced feature spaces. *Nature*, 567, 209–212.
<https://doi.org/10.1038/s41586-019-0980-2>
- PennyLane Development Team. (2020). PennyLane: Automatic differentiation of hybrid quantum-classical computations. *arXiv preprint*.
<https://doi.org/10.48550/arXiv.1811.04968>
- Orús, R., Mugel, S., & Lizaso, E. (2019). Quantum computing for finance: Overview and prospects. *Reviews in Physics*, 4, 100028.
<https://doi.org/10.1016/j.revip.2019.100028>
- Schuld, M. (2021). Supervised quantum machine learning models are kernel methods. *arXiv preprint*.<https://doi.org/10.48550/arXiv.2101.11020>
- Sezer, O. B., Gudelek, M. U., & Ozbayoglu, A. M. (2020). Financial time series forecasting with deep learning: A systematic literature review. *Applied Soft Computing*, 90, 106181.
<https://doi.org/10.1016/j.asoc.2020.106181>
- Fischer, T., & Krauss, C. (2018). Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2), 654–669.
<https://doi.org/10.1016/j.ejor.2017.11.054>
(Still extensively cited in 2020–2025 literature)
- Qin, Y., Song, D., Chen, H., Cheng, W., Jiang, G., & Cottrell, G. (2017). A dual-stage attention-based recurrent neural network for time series prediction. *Proceedings of IJCAI*.
<https://doi.org/10.24963/ijcai.2017/366>
(Key attention-based baseline referenced by later works)
- Borovykh, A., Bohte, S., & Oosterlee, C. W. (2018). Dilated convolutional neural networks for time series forecasting. *Journal of Computational Finance*, 22(4), 1–26.
<https://doi.org/10.21314/JCF.2018.360>
- Oreshkin, B. N., Carpow, D., Chapados, N., & Bengio, Y. (2020). N-BEATS: Neural basis expansion analysis for interpretable time series forecasting. *International Conference on Learning Representations (ICLR)*.
<https://doi.org/10.48550/arXiv.1905.10437>
- Salinas, D., Flunkert, V., Gasthaus, J., & Januschowski, T. (2020). DeepAR: Probabilistic forecasting with autoregressive recurrent networks. *International Journal of Forecasting*, 36(3), 1181–1191.
<https://doi.org/10.1016/j.ijforecast.2019.07.001>
- Rangapuram, S. S., Seeger, M., Gasthaus, J., et al. (2021). Deep state space models for time series forecasting. *Advances in Neural Information Processing Systems (NeurIPS)*.
<https://doi.org/10.48550/arXiv.1803.10122>
- Wen, Q., Zhou, T., Zhang, C., et al. (2023). Transformers in time series: A survey. *IEEE Transactions on Knowledge and Data Engineering*.
<https://doi.org/10.1109/TKDE.2023.3236455>
- Lim, B., & Zohren, S. (2021). Time-series forecasting with deep learning: A survey. *Philosophical Transactions of the Royal Society A*, 379(2194), 20200209.
<https://doi.org/10.1098/rsta.2020.0209>
- Siarni-Namini, S., Tavakoli, N., & Namin, A. S. (2018). A comparison of ARIMA and LSTM in forecasting time series. *2018 17th IEEE International Conference on Machine Learning and Applications*.
<https://doi.org/10.1109/ICMLA.2018.00227>
(Benchmark comparison frequently cited post-2020)
- Jiang, W. (2021). Applications of deep learning in stock market prediction: Recent progress. *Expert Systems with Applications*, 184, 115537.
<https://doi.org/10.1016/j.eswa.2021.115537>
- Patel, J., Shah, S., Thakkar, P., & Kotecha, K. (2015). Predicting stock and stock price index movement using trend deterministic data preparation and machine learning techniques. *Expert Systems with Applications*, 42(1), 259–268.
<https://doi.org/10.1016/j.eswa.2014.07.040>
(Classic feature-engineering reference still used in 2020+)

- Kim, K. J. (2003). Financial time series forecasting using support vector machines. *Neurocomputing*, 55(1–2), 307–319.
[https://doi.org/10.1016/S0925-2312\(03\)00372-2](https://doi.org/10.1016/S0925-2312(03)00372-2)
(Referenced to contrast DL approaches)
- Rebertus, P., Mohseni, M., & Lloyd, S. (2014). Quantum support vector machine for big data classification. *Physical Review Letters*, 113(13), 130503.
<https://doi.org/10.1103/PhysRevLett.113.130503>
(Foundation of quantum ML applied in later finance studies)
- Zoufal, C., Lucchi, A., & Woerner, S. (2019). Quantum generative adversarial networks for learning and loading random distributions. *npj Quantum Information*, 5, 103.
<https://doi.org/10.1038/s41534-019-0223-2>
- Schuld, M., & Petruccione, F. (2021). Machine learning with quantum computers. *Springer Nature*.
<https://doi.org/10.1007/978-3-030-83098-4>
- Abbas, A., Sutter, D., Zoufal, C., et al. (2021). The power of quantum neural networks. *Nature Computational Science*, 1, 403–409.
<https://doi.org/10.1038/s43588-021-00084-1>
- Du, Y., Hsieh, M.-H., Liu, T., & Tao, D. (2021). Expressive power of parametrized quantum circuits. *Physical Review Research*, 3(2), 023126.
<https://doi.org/10.1103/PhysRevResearch.3.023126>
- Holmes, Z., Sharma, K., Cerezo, M., & Coles, P. J. (2022). Connecting ansatz expressibility to gradient magnitudes and barren plateaus. *PRX Quantum*, 3(1), 010313.
<https://doi.org/10.1103/PRXQuantum.3.010313>
- Perdomo-Ortiz, A., Benedetti, M., Realpe-Gómez, J., & Biswas, R. (2018). Opportunities and challenges for quantum-assisted machine learning in near-term quantum computers. *Quantum Science and Technology*, 3(3), 030502.
<https://doi.org/10.1088/2058-9565/aab859>
- Mugel, S., Kuchkovsky, C., Sánchez, E., Fernández-Lorenzo, S., Luis-Hita, J., & Orús, R. (2022). Dynamic portfolio optimization with real datasets using quantum processors and quantum-inspired tensor networks. *Physical Review Research*, 4(1), 013006.
<https://doi.org/10.1103/PhysRevResearch.4.013006>
- Orús, R., Mugel, S., & Lizaso, E. (2019). Quantum computing for finance: Overview and prospects. *Reviews in Physics*, 4, 100028.
<https://doi.org/10.1016/j.revip.2019.100028>
- García-Martín, D., & Sierra, G. (2022). Quantum machine learning for finance: A review. *IEEE Access*, 10, 121091–121105.
<https://doi.org/10.1109/ACCESS.2022.3215946>
- Alodadi, A., & Rezgui, A. (2023). Stock price prediction using deep learning and technical indicators. *Applied Soft Computing*, 132, 109851.
<https://doi.org/10.1016/j.asoc.2022.109851>
- Patel, S., Doshi, N., & Shah, P. (2024). Multivariate deep learning models for stock price forecasting with technical indicators. *Neural Computing and Applications*.
<https://doi.org/10.1007/s00521-024-08671-9>
- Md. A. Quasem, M. Hossain, M. M. Rahman, and M. R. Islam, “An optimized multi-layer stacked LSTM model for stock price prediction,” *Expert Systems with Applications*, vol. 213, pp. 118899, 2023. doi: 10.1016/j.eswa.2022.118899.
- A. Liapis, C. Makris, and D. Skoutas, “Multivariate time-series forecasting using temporal convolutional networks with textual sentiment embeddings,” *Expert Systems with Applications*, vol. 212, pp. 118742, 2023. doi: 10.1016/j.eswa.2022.118742.
- X. Zhang, J. Gao, and Y. Li, “A comprehensive survey on temporal convolutional networks for time-series forecasting,” *IEEE Transactions on Neural Networks and Learning Systems*, 2023. doi: 10.1109/TNNLS.2023.3274124.
- T. Muhammad, A. B. Aftab, M. Ibrahim, M. M. Ahsan, M. M. Muhu, S. I. Khan, and M. S. Alam, “Transformer-based deep learning model for stock price prediction: A case study on Bangladesh stock market,” *International Journal of Computational Intelligence and Applications*, vol. 22, no. 03, 2023. doi: 10.1142/S146902682350013X.

- M. Doosti, M. Schuld, and N. Killoran, “Empirical evaluation of hybrid quantum–classical models on real-world regression tasks,” *Scientific Reports*, vol. 14, pp. 11873, 2024. doi: 10.1038/s41598-024-62591-7.